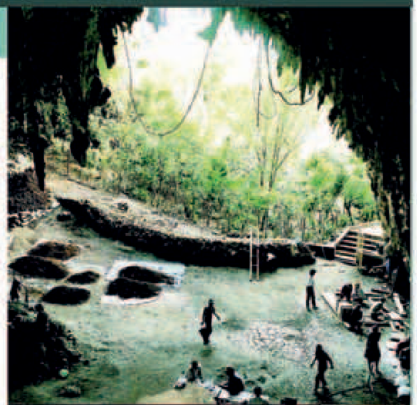
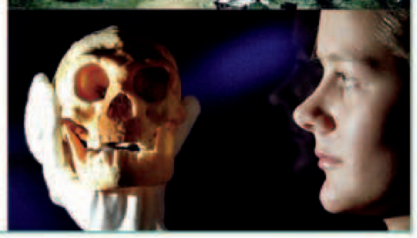
HOW WE KNOW

FLORES FIND

Excavations in 2003 at Liang Bua Cave (right) on Flores Island, in Indonesia, yielded the remains of a tiny skeleton standing about 3 ft 6 in (1 m) tall. The bones display a unique mix of primitive and more advanced characteristics, and date to about 18,000 years ago. With a small skull (below), large brow ridge, and a delicate face, *Homo floresiensis* had slight legs like some early hominins, yet modern teeth. Questions have been raised over whether this is a separate species or a small *Homo sapiens*. Others suggest this is the remnant of a *Homo erectus* population, or the descendant of humans who drifted to the island, then developed unique anatomical traits in isolation. Unless more remains are found, *Homo floresiensis* may remain an intriguing, unsolved mystery.





» deserts and settled in northeastern China by 25,000 years ago, after the warmer south part of the continent had been explored.

Sunda, Sahul, and Asia

During the late Ice Age, a huge continental shelf—an area of land connecting the continents that is now covered by higher sea levels—known as Sunda extended from mainland Southeast Asia far into the Pacific. Only short stretches of open water separated New Guinea and Australia from this now-sunken land. Another landmass, Sahul, linked Australia and New Guinea themselves. *Homo sapiens* arrived in mainland Southeast Asia before 50,000 years ago. By 45,000 years ago—the date is controversial—a few hunting bands had crossed open water to Sahul and colonized what is now Australia. They may have crossed on primitive rafts or in dugout canoes. Modern humans had settled New Guinea by about 40,000 years ago, and crossed to the Solomon Islands by about 5,000 years later. Hunter-gatherers had settled throughout Australia, including Tasmania, by 30,000 years ago.

This was the outer limit of human settlement of the offshore Pacific until outrigger canoes (see pp.216–17) and open-water navigation techniques allowed people with domesticated animals and root crops to make the lengthy open-water passages after 1000 BCE. The evidence of human life at Lake Mungo in Australia reveals details of hunter-gatherer life about 40,000 years ago. It is important as it captures a moment in time and a lifestyle that remained largely unchanged for thousands of years.

Reaching the Americas

Archaeologists have disputed the date of the first settlement of the Americas for over a century. Most now agree that native Americans originated in Siberia. Genetic and dental evidence links the two areas and backs up this theory. There are also linguistic ties that hint at population movements from Siberia to Alaska. But it is not known precisely when and how the first settlement took place.

Until about 10,000 years ago, a low-lying land bridge, Beringia, joined Siberia to Alaska (see pp.24–25). Most scientists believe that the first Americans were Siberian hunters who crossed this bridge into Alaska at least 15,000 years ago, toward the end of the Ice Age.

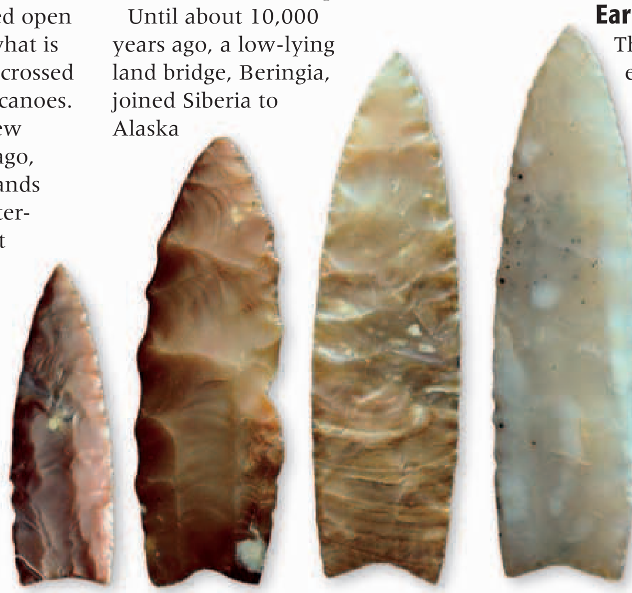
Route south

More controversy surrounds the route by which the first Americans penetrated the heart of North America, something which is thought to have taken place at least 13,000 years ago. Huge ice sheets covered most of what is now Canada. One theory favors a movement south along the continental shelves of southeast Alaska and British Columbia, which was then a landscape of steppe-tundra. Another common hypothesis claims a rapid movement south along a narrow corridor between two ice sheets, one mantling the Rocky Mountains and the other extending east toward the Atlantic. The controversy is unresolved, but we know that small numbers of early American hunter-gatherers were south of the ice sheets, and some as far south as Chile, by at least 13,000 years ago.

The early Americans are best known from the remains of kills of bison, mammoth, and mastodon in North America. They are often labeled "big-game" hunters, which is misleading, as they relied on plant foods and adapted to temperate and tropical areas, as well as the bleak lands at the margins of retreating ice sheets. They did prey on indigenous species of large mammals, but, by 10,500 years ago, most of this "megafauna" was extinct, probably as a result of drier climatic conditions, perhaps speeded by some overhunting.

Early evidence

The archaeological record of the early Americas is sketchy. Key sites include a 12,000-year-old rock shelter in Meadowcroft, Pennsylvania, a scatter of stone tools from a site at Cactus Hill, Virginia, and a well-documented foraging camp at Monte Verde, Chile, dating to about 13,000 years ago. The first well-defined culture is that of the Clovis people, famous for their fine flint tools, who flourished between about 11,200 and 10,900 years ago. One controversial discovery is a 9,500 year-old skull from Kennewick, Washington State, which is believed to have caucasian features and may be an indication that some of the first settlers in America came from Europe. However, this has been the subject of much debate.



Oldest footprints
Hundreds of human footprints, preserved for over 20,000 years, have been found at Lake Mungo, Australia. At that time, the lake there would have been home to fish, mussels, and crayfish—all valuable food sources.

Clovis points
North American hunters made these flint spearpoints over 11,000 years ago. They are some of the few objects found from this early period. They would have been used to kill and cut up large prey such as mammoth.

AFTER

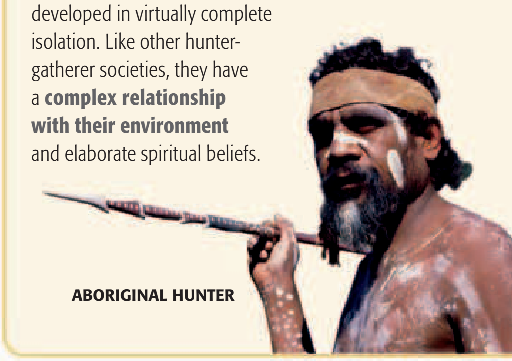
By 10,000 years ago humans had spread to every continent (except Antarctica) and had learned the skills needed to survive in different environments. Later explorers found their "new world" already inhabited by the descendants of those first settlers.

ADAPTING TO CHANGE
American Indian societies adjusted to warmer, often drier conditions, by intensifying the search for food, whether it be fish, game, or plant foods. By 4000 BCE, some foraging groups were experimenting with the **planting of native grasses 36–37 »**, such as goosefoot.

EUROPEAN SETTLERS IN AMERICA

LATER EXPLORATION
Europeans first came in **contact with American Indians** 500 years ago when they traveled the world in search of new land **230 »**. Dutch settlers arrived in Manhattan in the 1800s and traded with the native population before establishing a permanent settlement there.

AN ISOLATED CULTURE
The culture of the Australian Aboriginals developed in virtually complete isolation. Like other hunter-gatherer societies, they have a **complex relationship with their environment** and elaborate spiritual beliefs.



ABORIGINAL HUNTER

INVENTION

ATLATL

Atlatls (from an Aztec word) are throwing sticks or spear-throwers, first developed by Cro-Magnon hunters over 20,000 years ago. Spear throwers increase a spear's range and velocity—useful qualities for hunters who rely on stalking to kill their prey. The simplest atlatls are hooked sticks. A weight adds stability and velocity to the throw. Such weights, often called "bannerstones," are often found on native American sites, as they arrived with the first inhabitants of the region. The Aztecs later used them against Spanish *conquistadors* (see pp.230–31).